

Ch # 3

Linear Programming.

Linear Programming may be defined as a method of determining an optimal program of inter dependence activities in view of available resources.

The object in a linear programming is to maximize profit or minimize loss, as the case may be, subject to a number of limitations known as constraints.

The solution to a linear programming shows how much would be produced valid (as sold or purchased) which will optimized the objective function and satisfied the constraints.

Ingredients of linear Programming:-

There are 4 ingredients of linear programming:

- ⇒ Decision Variable.
- ⇒ Constraints.
- ⇒ Parameters / data.
- ⇒ Objective functions.

Concept of primal & dual:

The basic linear programming is called primal or primal program, while its counter part is called Dual or Dual programme.

In this situation, the maxima and minima would not be representing two difference equations. The fact is that optimal values in the primal of objective function and the optimal values in the dual of objective

function are always equal.

"In mathematical optimization theory, duality or the duality principle is the principle that optimization problems may be viewed from either of two perspective, the primal problem or dual problem if the primal is a minimization problem then the dual is a maximization problem and vice versa."

Duality theorems:-

Optimal value of primal and dual identical as

$$\bar{\pi} = \bar{\pi}^*, \quad \bar{c} = \bar{c}^*$$

The two dual theorems are of great importance

- 1- The optimal value of the primal objective function always equals the optimal value of the dual objectives solution exists.
- 2- If choice variables of a programme is non-zero. Then corresponding dummy variable is non-zero & vice versa

Primal dummy variable is denoted by s_i .
Dual dummy variable is denoted by t_j .

$$\text{Primal } \begin{cases} x_i > 0 \Rightarrow t_j = 0 \\ x_i = 0 \Rightarrow t_j > 0 \end{cases}$$

$$\text{Dual } \begin{cases} y_j > 0 \Rightarrow s_i = 0 \\ y_j = 0 \Rightarrow s_i > 0 \end{cases}$$

Q:-What is Duality?

Ans: Corresponding to every minimization programme (To minimize C). There always exist a counter part maximization (To minimize a new variable C^*) with the property.

$$\bar{C} = \bar{C}^*$$

Similarly for every maximization programme (To maximize π) There, ^{always} exist a minimization programme (To minimize a new variable π^*) with the property

$$\bar{\pi} = \bar{\pi}^*$$

The original programme is known as primal and the counter programme is known as dual.

As the optimal values of both primal and dual are the same. So we have a choice to select easier one.

Rules of Transformations:-

1. Change maximization to minimization and vice versa. and attach "Asterisk" to the dual programme.
2. For the co-efficient matrix of a dual programme take the transpose of the co-efficient matrix of the primal programme.
3. Row vector in the primal programme will become column vector in the dual programme and vice versa.
4. Inequality signs must be changed. Although non-negative restriction will not be changed.

Simplex Methods-

Under this method linear programming is used to find optimal solution or to make optimality test. Under this method the $n-m$ variables are kept equal to zero in each stage till " m equations are solved for n variables. The purpose of this method is a movement from one feasible solution to another feasible solution. The mathematical operation made under simplex method is based upon matrix algebra.

Slack variables:-

are those variables which are used in the system of linear inequalities must be converted to a system of linear equations. For Example: $5x_1 + 3x_2 \leq 30$ can be converted to an equation by adding a slack variable $S \geq 0$ such that $5x_1 + 3x_2 + S = 30$.

Surplus Variables:-

are those variables which are used to convert an equation by subtracting a surplus variable. Such that $4x_1 + 7x_2 - S = 60$.
Hence surplus variables are subtracted variable i.e.,

$$2y_1 + y_2 - S_1 = 14, \quad y_1 + y_2 - S_2 = 12$$

$$y_1 + 3y_2 - S_3 = 18.$$

Feasible region:-

All points do satisfy all three constraints simultaneously is called feasible region.

Linear programming– general

- * Many economic problems involve the optimization of a certain objective (e.g., profits) subject to restrictions (e.g., production function)
- * Linear programming is an application of optimization that is frequently applied in many decision-making situations, such production, inventory management, planning, scheduling, and so on
- * In linear programming, both the objective function and the constraints are linear
- * The feasible solution satisfies all the constraints while maximizing (or minimizing) the objective function



Managerial Economics Linear Programming

Slides:

Linear programming– general

- Understand the problem: what is the objective and what is (are) the constraint(s)
- * Write the objective in terms of the decision variables
- * Write the constraints in terms of the decision variable
- * Example 1: maximization problem

The company offers two training programs, one of them last two days on team building and the other lasts three days on problem solving. Company management wants to offer at most 6 training programs on teaming during the next 2 months. At most 8 programs are offered during the period. A consultant providing training is paid for 19 days.

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Managerial Economics Linear Programming

Slides:

Linear programming– general

- * Sensitivity analysis is a way to evaluate the robustness of the solution
- * This could involve sensitivity of the solution to changes in the coefficients (which may be subject to variation) or the value of the constraints
- * It will allow a manager to have a better understanding of constraints and limits of the problem
- * E.g., the slope of the objective function is different or the budget allows additional days for consultant work

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Slides:

Linear programming– examples

- * Read the following two papers:
 - * Example 1: Linear Programming Applications (discussed in class if time permits)
 - * Example 2: Introduction to Linear Programming



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Slides:

Linear programming– general

- What is the feasible region?
- Draw the constraints and the objective function into the graph (axis are x_1 and x_2)
- What is the optimal solution?
- Please note that not always there is a unique optimal solution



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Slides:

Linear programming– general

Training program on team building is estimated to bring in \$5 (x1,000) and on problem solving \$7 (x1,000).

* Formulate the problem:

Let x_1 = number of training programs on team building
and x_2 = number of training programs of problem solving

Max the value of $V = 5x_1 + 7x_2$ subject to

$$x_1 \leq 6$$

$$x_1 + x_2 \leq 8$$

$$2x_1 + 3x_2 \leq 19$$

$$x_1 \geq 0, x_2 \geq 0$$



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Viewed as a resource allocation model, the primal problem has n economic activities and m resources. The coefficient C_j in the primal represents the revenue per unit of activity j . Resource i , whose maximum availability is b_i , is consumed at the rate a_{ij} units per unit of activity j .

ECONOMIC INTERPRETATION OF DUALITY

The linear programming problem can be viewed as a resource allocation model in which the objective is to maximize revenue subject to the availability of limited re-sources. Looking at the problem from this standpoint, the associated dual problem of-fers interesting economic interpretations of the LP resource allocation model.

To formalize the discussion, we consider the following representation of the general primal and dual problems:

Primal	Dual
$\text{Maximize } z = \sum_{i=1}^n c_j x_j$ subject to $\sum_{j=1}^n a_{ij} x_j \leq b_i, i = 1, 2, \dots, m$ $x_j \geq 0, j=1, 2, \dots, n$	$\text{Minimize } w = \sum_{i=1}^m b_i y_i$ subject to $\sum_{i=1}^m a_{ij} y_i \geq c_j, j = 1, 2, \dots, n$ $y_i \geq 0, i = 1, 2, \dots, m$